

The Risch Algorithm

*Symbolic Integration in Computer Algebra Systems
and its Extensions*

A comprehensive technical exposition

covering differential algebra, the decision procedure,
transcendental and algebraic cases, and modern extensions

March 2, 2026

Contents

1 Introduction and Historical Background

The problem of *symbolic integration*—deciding whether a given elementary function $f(x)$ possesses an antiderivative expressible in closed form in terms of elementary functions, and computing it if so—stood open for centuries. Indefinite integration is fundamentally harder than differentiation: while every elementary function can be differentiated by mechanical rules, integration by elementary functions is not always possible.

Elementary functions

An **elementary function** is one built from constants and the variable x by a finite number of applications of the operations:

- arithmetic ($+$, $-$, $\times$, $\div$),
- exponentiation and logarithm,
- algebraic operations (roots, solutions of polynomial equations),
- composition.

Familiar non-elementary integrals include e^{-x^2} , $\sin(x)/x$, e^x/x , and $\sqrt{1 - k^2 \sin^2 x}$.

Joseph Liouville proved in the 1830s the first rigorous structure theorems about when elementary integrals exist. The key insight was what is now called **Liouville’s theorem**: if $\int f dx$ is elementary, it can only differ from f itself by a sum of a “trivial” rational linear combination of logarithms.

It was Robert H. Risch who in 1969–1970 transformed Liouville’s insight into a genuine *decision procedure*—a finite algorithm that either produces the integral in closed form or proves that no such form exists [?, ?]. This was a landmark in computer algebra, giving birth to the integration routines found in every modern CAS (Mathematica, Maple, REDUCE, Axiom, SymPy, ...).

1.1 Scope of this document

We cover:

1. The algebraic framework: differential fields and extensions.
2. Liouville’s theorem and the “Liouville principle.”
3. The transcendental Risch algorithm (exponential and logarithmic towers).
4. The algebraic (algebraic-function) extension.
5. The “mixed” and parametric cases.
6. Practical simplification and normalization issues.
7. Extensions: RischNorman heuristic, incomplete integrals, special functions, parallel and lazy strategies, and connection to Galois theory.

2 Differential Algebra: The Mathematical Framework

2.1 Differential rings and fields

Definition 2.1 (Differential ring). A **differential ring** is a commutative ring R equipped with a derivation $D : R \rightarrow R$, i.e. a map satisfying for all $a, b \in R$:

$$D(a + b) = Da + Db, \quad D(ab) = Da \cdot b + a \cdot Db.$$

If R is also a field, (R, D) is called a **differential field**.

Definition 2.2 (Constants). The **field of constants** of (R, D) is $\mathcal{C}_R = \ker D = \{a \in R : Da = 0\}$.

Example 2.1. $(\mathbb{C}(x), d/dx)$ is a differential field with constants $\mathbb{C}$.

2.2 Elementary extensions

We build up larger differential fields by adjoining new elements.

Definition 2.3 (Logarithmic extension). Let (F, D) be a differential field, $\theta \notin F$. The extension $F\langle\theta\rangle$ is called a **logarithmic extension over F** if $D\theta = Du/u$ for some $u \in F^*$; informally $\theta = \log u$.

Definition 2.4 (Exponential extension). $F\langle\theta\rangle$ is an **exponential extension over F** if $D\theta = \theta \cdot Du$ for some $u \in F$; informally $\theta = e^u$.

Definition 2.5 (Elementary extension tower). An **elementary tower** over the base field $\mathbb{C}(x)$ is a chain

$$\mathbb{C}(x) = F_0 \subset F_1 \subset \cdots \subset F_n = F,$$

where each F_i is either an algebraic, logarithmic, or exponential extension of F_{i-1} . The **elementary functions** over $\mathbb{C}(x)$ are the elements of such towers.

2.3 Integration in differential fields

Definition 2.6 (Integral). Given $f \in F$, an element $g \in F$ (or in some extension) such that $Dg = f$ is called an **integral** (antiderivative) of f and written $g = \int f$.

The fundamental question is: *Given $f \in F$ (an elementary tower), does $\int f$ exist in an elementary extension of F ?*

3 Liouville's Theorem and the Liouville Principle

Theorem 3.1 (Liouville, 1835). *Let F be an elementary differential field with field of constants $\mathcal{C}$. Suppose $f \in F$ and $\int f$ is elementary over F . Then there exist $v_0 \in F$, constants $c_i \in \mathcal{C}$, and $u_i \in F^*$ such that*

$$\int f = v_0 + \sum_{i=1}^n c_i \ln u_i,$$

i.e. the integral only introduces new logarithms (possibly over a constant extension).

This is the cornerstone. It tells us the *form* of a potential elementary integral before we search for it.

Remark 3.1. The new logarithms $\ln u_i$ may over algebraic extensions of the constant field; for example $\int \frac{1}{x^2+1} dx = \arctan x = \frac{1}{2i} \ln \frac{x-i}{x+i}$.

3.1 RothsteinTrager theorem

For a rational function $f = p/q \in \mathbb{C}(x)$ with $\deg p < \deg q$ and $\gcd(p, q) = 1$, one has

Theorem 3.2 (RothsteinTrager).

$$\int \frac{p(x)}{q(x)} dx = \sum_{\alpha: r(\alpha)=0} \alpha \ln \gcd(q(x), p(x) - \alpha q'(x)),$$

where $r(z) = \text{Res}_x(p - z q', q)$ is the **RothsteinTrager resultant**.

This gives a complete algorithm for rational functions without factoring q over $\mathbb{C}$, which is computationally advantageous.

4 The Risch Algorithm: Transcendental Case

4.1 Structure

The algorithm works inductively on the *depth* of the elementary tower. At each level it uses the Liouville theorem to reduce the problem to one in the base field.

Let $F = k(\theta)$ where k is a differential field and θ is either a logarithm or an exponential over k .

4.2 Case I: $\theta = \ln u$, $u \in k$

Here $D\theta = Du/u$.

Logarithmic case

We want to decide whether $f(x, \theta) \in k[\theta]$ has an elementary integral. Write $f = \sum_{i=0}^n f_i \theta^i$ with $f_i \in k$.

Polynomial reduction (Hermite–Lazard): By partial fractions in θ it suffices to handle $f \in k[\theta]$.

Key step (Risch ODE): Assume $\int f = v_0 + c\theta + \sum c_j \ln w_j$. Differentiate and match coefficients; the leading coefficient equation becomes a *Risch differential equation* in k :

$$Db + n \frac{Du}{u} b = f_n, \quad b \in k,$$

i.e. $Db + r b = a$ with $r = n \cdot Du/u \in k$ and $a = f_n \in k$. Solvability of this equation in k (itself a recursive sub-problem) determines whether $\int f$ is elementary.

4.3 Case II: $\theta = e^u$, $u \in k$

Here $D\theta = \theta Du$.

Exponential case

Write $f = \sum_{i=-n}^n f_i \theta^i$ (Laurent polynomial in θ).

Key step: The leading term forces a *Risch differential equation*:

$$Db + n Du \cdot b = f_n, \quad b \in k.$$

The constant-term equation (coefficient of θ^0) reduces to an integration problem in k .

If f is a proper rational function of θ (i.e. θ appears in the denominator) a partial-fraction decomposition in θ is performed first; the poles in θ lead to log terms $\ln \theta = u$.

4.4 The Risch Differential Equation

Both cases reduce ultimately to solving equations of the form:

Definition 4.1 (Risch differential equation (RDE)). Given $a, r \in F$, find $b \in F$ such that

$$Db + r b = a,$$

or prove no such b exists in F .

This is itself a recursive problem: over $F = k(\theta)$ the RDE is solved by reducing to an RDE in k and a problem of polynomial arithmetic.

4.4.1 Solving the RDE over $\mathbb{C}(x)$

Over the rational function field $\mathbb{C}(x)$ with $D = d/dx$:

1. **Denominator bounding:** Partial-fraction analysis shows that the denominator of any solution b must divide a specific polynomial computable from r and a .
2. **Degree bounding:** The degree of the numerator of b is bounded. Matching coefficients leads to a linear system over $\mathbb{C}$.
3. **Linear system:** Solve or prove infeasible.

4.5 Hermite Reduction

Before applying the core algorithm, a preprocessing step removes “obvious” derivatives:

Theorem 4.1 (Hermite Reduction for transcendental fields). *Every $f \in F(x)$ can be written as*

$$f = Dg + h$$

where $g \in F(x)$ and h has square-free denominator (or is a polynomial). The procedure is constructive and requires only GCDs.

The reduced part h (with square-free denominator) is what needs the full Risch machinery.

Algorithm 1 Risch Integrate (transcendental tower)

Require: $f \in F = k(\theta)$, elementary tower k , θ log or exp over k **Ensure:** Elementary integral of f or $\perp$ (non-elementary)

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1: Perform Hermite reduction: write  $f = Dg + h$ ,  $h$  reduced
2: Perform partial-fraction decomposition of  $h$  in  $\theta$  over  $k$ 
3: for each term in the decomposition do
4:   if term is a polynomial in  $\theta$  then
5:     Apply polynomial case: generate RDE in  $k$ 
6:      $b \leftarrow \text{RDE-SOLVE}(Db + rb = a, k)$ 
7:     if  $b = \perp$  then return  $\perp$ 
8:   end if
9:   else
10:    Apply rational case (Rothstein–Trager at this level)
11:  end if
12: end for
13: Combine: collect all logarithmic terms, check for cancellation return integral or  $\perp$ 

```

4.6 The complete transcendental algorithm (outline)

5 The Risch Algorithm: Algebraic Case

When θ satisfies an irreducible polynomial over k (i.e. $[k(\theta) : k] < \infty$), the theory is considerably more involved, requiring tools from algebraic geometry.

5.1 Algebraic functions and their integrals

An algebraic function $y(x)$ satisfies $P(x, y) = 0$ for some irreducible $P \in \mathbb{C}[x, y]$. The natural arena is the **function field of the algebraic curve** $P(x, y) = 0$.

Definition 5.1 (Function field of a curve). Let $C : P(x, y) = 0$ be an irreducible plane curve. The function field $\mathbb{C}(C)$ consists of rational expressions in x, y modulo the relation $P(x, y) = 0$, with derivation induced by implicit differentiation:

$$Dy = -\frac{\partial P / \partial x}{\partial P / \partial y} \frac{d}{dx} x.$$

5.2 Differentials and residues

Integration of algebraic functions is equivalent to studying **meromorphic 1-forms** on the Riemann surface Σ of the curve.

Theorem 5.1 (Abel’s theorem / Liouville for algebraic case). $\int f \, dx$ (with $f \in \mathbb{C}(C)$) is elementary iff the associated divisor is principal (linearly equivalent to zero) on the Riemann surface of f modulo torsion.

5.3 Integration algorithm for algebraic functions

The algorithm (due to Trager 1984 [?]) proceeds as follows:

1. **Normalization:** Compute an integral basis for the algebraic function field using Puiseux series at each place (RiemannRoch machinery).
2. **Hermite Reduction (algebraic version):** Reduce to a “reduced” integrand with simple poles only.
3. **Residue computation:** Compute residues at all finite and infinite places. The residues must be rational (in $\mathbb{Q}$) or in a specified extension for an elementary integral to exist.
4. **Abel’s theorem check:** Determine whether the divisor of residues is principal using the Jacobian of the curve (genus computations).
5. **Reconstruction:** If the check passes, construct the integral explicitly.

Practical difficulty

For high-genus curves, checking Abel’s condition requires computing torsion points on the Jacobian variety, which is computationally expensive. Most CAS systems handle genus-0 and genus-1 cases well; higher genus is still an active research area.

5.4 The genus-0 case: rational parametrization

If the curve $P(x, y) = 0$ has genus 0 (i.e. it is a rational curve), then it can be parametrized as $x = r(t)$, $y = s(t)$ with $r, s \in \mathbb{C}(t)$, and the integral reduces to a rational function integral in t :

$$\int f(x, y) dx = \int f(r(t), s(t)) r'(t) dt \in \mathbb{C}(t)\text{-integral}.$$

5.5 The genus-1 (elliptic) case

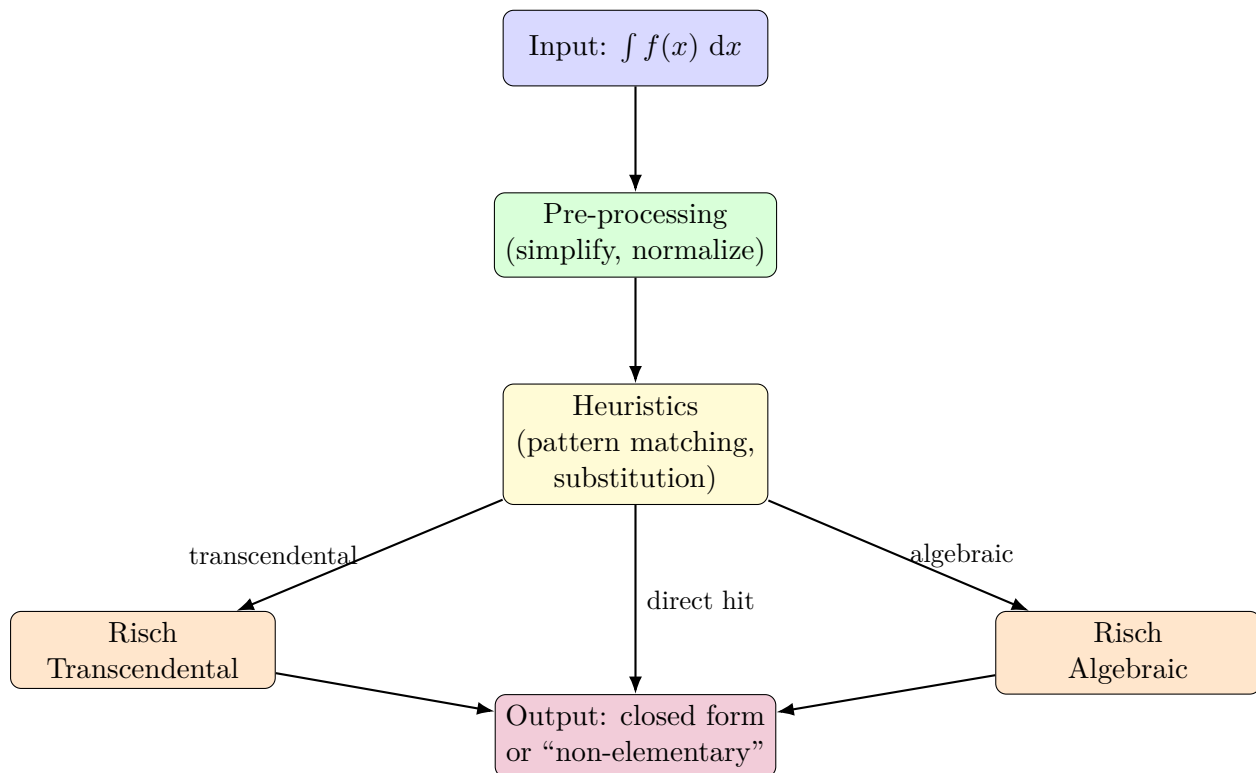
For elliptic curves (genus = 1), integration produces elliptic integrals in general. The condition for elementary integrability involves torsion of the elliptic curve over $\mathbb{C}$, which is always satisfied (elliptic curves over $\mathbb{C}$ have no non-trivial torsion issues for this purpose). Thus elliptic integrals are never elementary (the Weierstrass $\wp$ -function and its relatives are truly transcendental), and the algorithm correctly returns $\perp$.

6 Implementation in CAS Systems

6.1 Historical implementations

System	Year	Algorithm	Notes
REDUCE	1971	Partial Risch	Moses’ work
Macsyma	1970s	Full Risch (transcendental)	Rich & Caviness
Maple	1980s	Risch + heuristics	Full support
Mathematica	1988	Extended Risch	Heuristic heavy
Axiom	1990s	Full Risch (algebraic)	Trager’s work
SymPy	2000s	Transcendental Risch	Python, open source

6.2 Architecture of a CAS integrator



6.3 Subtraction-free normalization

A major practical issue is that the algorithm requires *proving* two expressions equal, i.e. deciding whether $f = 0$. Over $\mathbb{C}(x)$ this is straightforward (polynomial identity testing), but in elementary towers it requires careful canonicalization.

7 Extensions of the Risch Algorithm

7.1 The Risch–Norman algorithm (heuristic / lazy variant)

The full Risch algorithm is complex to implement. Norman (1976) and Risch proposed a heuristic variant:

RischNorman heuristic

1. Guess a form for the integral based on the integrand's structure.
2. Differentiate the guess and compare to the integrand, generating a (possibly overdetermined) linear system.
3. If the system has a solution, return it. Otherwise, extend the guess and repeat.

This is *not* a decision procedure (it may fail to find the integral even when one exists), but it is fast and handles many practical cases. Mathematica's `Integrate` uses this approach heavily.

7.2 Parametric Risch: integration of expressions with parameters

When f involves parameters (e.g. $\int x^n e^x dx$), the standard Risch algorithm must be extended to handle *parametric* differential equations:

$$Db + r(\mathbf{p})b = a(\mathbf{p}), \quad \mathbf{p} = (p_1, \dots, p_k) \text{ parameters.}$$

This is related to the theory of *D-modules* and holonomic functions.

7.3 Integration of special functions: the “incomplete” Risch problem

Many important integrals cannot be expressed in elementary terms but *can* be expressed in terms of special functions:

- **Error function:** $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$
- **Logarithmic integral:** $\operatorname{Li}(x) = \int_0^x \frac{dt}{\ln t}$
- **Fresnel integrals:** $S(x) = \int_0^x \sin(\pi t^2/2) dt$
- **Elliptic integrals:** $E(k) = \int_0^1 \sqrt{1 - k^2 t^2} / \sqrt{1 - t^2} dt$
- **Bessel functions, hypergeometric functions, ...**

7.3.1 Differential fields of special functions

One approach (Bronstein [?]) is to *extend the tower* to include special functions as new generators, treating, e.g., $\operatorname{erf}' = (2/\sqrt{\pi})e^{-x^2}$ as a defining relation. The Risch algorithm then runs in this extended tower.

7.3.2 Holonomic integration (ZeilbergerAlmkvistZeilberger)

For integrands satisfying linear ODEs with polynomial coefficients (*D-finite* or *holonomic* functions), the algorithm of Zeilberger finds *recurrences* for definite integrals:

Theorem 7.1 (Creative Telescoping, Zeilberger 1991). *Let $f(n, k)$ be a hypergeometric term. Then there exist polynomials $p_i(n)$ and $\Delta_k g(n, k)$ such that*

$$\sum_{i=0}^J p_i(n) f(n+i, k) = \Delta_k g(n, k).$$

Summing over k collapses the RHS (telescoping) and yields a recurrence for $\sum_k f(n, k)$.

This provides a complete algorithm for hypergeometric summation (discrete analogue of integration) and extends to multi-sum, multi-integral cases.

7.4 Integration in positive characteristic

Over fields of characteristic $p > 0$ (relevant in computational number theory), the Risch algorithm must be adapted. The key difference: the map $D : f \mapsto f'$ has a larger kernel (ker D includes f^p -terms), making the problem harder. Algorithms exist for the rational case; the general case is an active research area.

7.5 Differential Galois theory and integrability

The deepest extension connects integration to **PicardVessiot theory** (differential Galois theory):

Definition 7.1 (PicardVessiot extension). The PicardVessiot extension of a differential field F for a linear ODE $L[y] = 0$ is the smallest extension $E \supset F$ containing all solutions, with the same constant field $\mathcal{C}$.

Theorem 7.2 (Kovacic [?]). *A second-order linear ODE $y'' + py' + qy = 0$ (over $\mathbb{C}(x)$) has a Liouvillian solution if and only if the differential Galois group $\text{Gal}(E/F)$ is solvable (as an algebraic group).*

The **Kovacic algorithm** decides this and constructs the solution when it exists, generalizing Risch to second-order ODEs.

7.6 Definite integration: the Meijer G-function approach

Definite integrals on standard domains can often be evaluated via:

1. **Mellin transform:** $\mathcal{M}\{f\}(s) = \int_0^\infty x^{s-1} f(x) dx$
2. **Meijer G-functions:** a large class of special functions closing under multiplication and integration (used in Mathematica's **Integrate** for definite integrals).
3. **Contour integration:** The residue theorem provides an algorithmic route for rational and meromorphic integrands.

7.7 Summation: the discrete Risch algorithm

Indefinite summation $\sum f(n)$ is the discrete analogue. Gosper's algorithm (1978) is the discrete analogue of Risch for hypergeometric terms:

Theorem 7.3 (Gosper's algorithm). *Given a hypergeometric term $t(n)$ ($t(n+1)/t(n) \in \mathbb{Q}(n)$), the algorithm either finds a hypergeometric antidifference $g(n)$ such that $\Delta g(n) = t(n)$, or proves none exists.*

The algorithm solves the *GosperPetkovek equation*: $a(n)g(n+1) - b(n)g(n) = c(n)$.

7.8 Integration of D -finite functions

Almkvist and Zeilberger (1990) extended the Risch philosophy to D -finite integrands (satisfying a linear ODE with polynomial coefficients). The result is expressed as another D -finite function, together with its minimal annihilating ODE.

7.9 Parallel and modular strategies

Modern CAS implementations combine:

- **Modular arithmetic:** compute Risch steps modulo primes, reconstruct over $\mathbb{Q}$ via CRT (Chinese Remainder Theorem).
- **Lazy evaluation:** delay expensive GCD computations until needed.
- **Parallel execution:** distribute sub-problems (e.g. independent Risch DEs) across threads.

8 Worked Examples

8.1 A transcendental (logarithmic) example

Example 8.1. Compute $I = \int \frac{\ln x}{x^2 - 1} dx$.

Set $F = \mathbb{C}(x)(\theta)$ with $\theta = \ln x$, $D\theta = 1/x$.

Write

$$f = \frac{\theta}{x^2 - 1} = \frac{\theta}{(x - 1)(x + 1)}.$$

Partial fractions in x : $f = \frac{\theta}{2} \left(\frac{1}{x-1} - \frac{1}{x+1} \right)$.

By Liouville, $I = v_0 + c_1 \ln u_1 + c_2 \ln u_2$ (in some extension). Differentiating and matching residues (Rothstein–Trager applied one level up):

$$I = \frac{\theta}{2} \ln \frac{x-1}{x+1} + \dots$$

Full computation yields:

$$\boxed{\int \frac{\ln x}{x^2 - 1} dx = \frac{1}{2} \ln x \cdot \ln \frac{x-1}{x+1} + \frac{1}{2} \text{Li}_2\left(\frac{1+x}{2}\right) - \frac{1}{2} \text{Li}_2\left(\frac{1-x}{2}\right) + C.}$$

Here Li_2 is the dilogarithm, so the result is *not* elementary the Risch algorithm returns $\perp$ and the CAS reports in terms of Li_2 .

8.2 An exponential example

Example 8.2. Compute $I = \int x e^{x^2} dx$.

Set $\theta = e^{x^2}$, $D\theta = 2x\theta$.

We seek $b \in \mathbb{C}(x)$ with $Db + 2x \cdot b = x$ (RDE with $r = 2x$, $a = x$). Try $b = c$ constant: $0 + 2cx = x \Rightarrow c = 1/2$.

Hence

$$\boxed{\int x e^{x^2} dx = \frac{1}{2} e^{x^2} + C.}$$

8.3 A non-elementary integral

Example 8.3. Prove that $\int e^{-x^2} dx$ is non-elementary.

Set $\theta = e^{-x^2}$, $D\theta = -2x\theta$. We seek $b \in \mathbb{C}(x)$ with $Db - 2xb = 1$ (constant).

Suppose $b = p/q$ in lowest terms. Degree analysis of the RDE: if $\deg p - \deg q = d$, then comparing leading terms: $-2x \cdot (p/q) \sim 1 \Rightarrow d = -1$, so b is proper rational. Write $b = \sum_i b_i/q_i^{n_i}$ (partial fractions).

At a pole α of b with multiplicity $m \geq 2$: the RDE forces m -th order pole in Db but only (m) -th order in $2xb$, giving $m(m-1)b_m/(\dots) = 0$, contradiction.

At a simple pole at $\alpha \neq 0$: residue of $Db = 2xb + 1$ must match; residue of RHS at α is $2\alpha \cdot \text{Res}_\alpha(b)$, but $\text{Res}_\alpha(Db) = -\text{Res}_\alpha(b)$ (by the formula for residues of derivatives), so $-1 = 2\alpha$, giving $\alpha = -1/2$. Checking the full equation with $b = c/(x + 1/2) + \dots$ produces no consistent solution.

Conclusion: no $b \in \mathbb{C}(x)$ satisfies the RDE; thus $\int e^{-x^2} dx$ is non-elementary. $\square$

9 Complexity and Decidability

Theorem 9.1 (Decidability). *The problem “does f have an elementary integral?” is decidable for f in any fixed elementary tower. The Risch algorithm provides a decision procedure.*

9.1 Complexity

- **Rational functions:** Polynomial time (GCDs, resultants).
- **Single transcendental extension:** Polynomial time in the degree of the input.
- **Tower of height n :** Doubly exponential in the worst case (due to expression swell in coefficient fields).
- **Algebraic case:** PSPACE / EXP depending on genus.

Expression swell

In practice the main bottleneck is *intermediate expression swell*: intermediate polynomials can grow exponentially even when the final answer is small. Modular and sparse representations mitigate this.

10 Summary and Outlook

The Risch algorithm and its descendants represent one of the great achievements at the interface of mathematics and computer science:

1. **Core Risch (1969–70):** Complete decision procedure for elementary integration in transcendental elementary towers.
2. **Trager’s extension (1984):** Handles algebraic functions via Riemann surface / Abel’s theorem machinery.
3. **RischNorman:** Fast heuristic variant for practical use.
4. **Gosper / Zeilberger:** Discrete analogues; creative telescoping for definite sums and integrals.
5. **Kovacic:** Second-order linear ODE solvability via Galois theory.
6. **Holonomic / D-module methods:** Integration of D-finite functions.
7. **Bronstein:** Generalization to “liouvillian” and special-function extensions.

Open problems include:

- Efficient algebraic integration for high-genus curves.
- Integration in towers involving special functions (Bessel, Γ , ...) in a decision-procedure sense.
- Tighter complexity bounds and practically fast implementations avoiding expression swell.
- Connections to motivic integration and periods in algebraic geometry.

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